CAPSTONE PROJECT PROPOSAL (Draft v2.0)

Project Title
SentinelAI: An AI-based Autonomous Security Agent for Threat Detection and Response in SOC and Data Center Environments

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1. Introduction and Background

1.1 Problem Description

Modern organizations rely heavily on Security Operation Centers (SOC) to monitor and respond to cybersecurity threats. SOC platforms collect massive amounts of security logs from sources such as SIEM systems, intrusion detection systems (IDS/IPS), endpoint detection and response tools (EDR), and cloud monitoring platforms.

However, the number of alerts generated by these systems continues to increase rapidly. Studies show that a large proportion of security alerts are false positives or low-priority events, forcing analysts to spend significant time investigating irrelevant alerts instead of focusing on real threats.

In large infrastructures such as data centers, attacks often involve multiple stages including reconnaissance, privilege escalation, lateral movement, and data exfiltration. These complex attack chains are difficult to identify using traditional rule-based detection systems.

Existing SOC automation solutions typically rely on static rules and predefined playbooks. While these methods help automate certain tasks, they lack the ability to perform contextual reasoning or adapt to evolving attack patterns.

Recent research suggests that artificial intelligence techniques such as machine learning and behavior analysis can significantly improve threat detection and security automation [1]. However, many existing AI-based solutions focus primarily on anomaly detection and do not provide comprehensive decision support for SOC analysts.

Therefore, there is a need for an intelligent system that can observe security events, correlate alerts into meaningful incidents, reason about potential threats, and assist analysts in responding to attacks more effectively.

1.2 Research Objectives

Main Objective

The main objective of this project is to design and develop an AI-based autonomous security agent that assists SOC analysts in detecting and responding to cybersecurity threats in data center environments.

Specific Objectives

• Analyze current approaches used in SOC automation and AI-based threat detection systems.
• Design an AI-based agent capable of correlating security events and identifying suspicious patterns.
• Develop a prototype system that assists analysts in analyzing threats and recommending responses.
• Evaluate the effectiveness of the proposed system in improving incident detection and response efficiency.

Research Questions

1. What existing methods are used in SOC automation and threat detection systems, and what are their strengths and limitations?
2. How can an AI-based security agent be designed to improve threat detection and response processes in SOC environments?
3. How can the performance of the proposed system be evaluated using measurable metrics such as detection accuracy and alert reduction?

1.3 Scope of the Project

Included

• AI-based analysis of security alerts
• Event correlation and incident detection
• Behavioral analysis of suspicious activities
• Decision-support recommendations for SOC analysts
• Integration with simulated SOC log data

Excluded

• Full enterprise-scale SOC deployment
• Real-time protection for national infrastructure systems
• Offensive penetration testing activities

The project focuses on developing a **prototype system** that demonstrates the feasibility of AI-assisted threat detection and response.

2. Proposed Solution

2.1 Solution Description

The proposed system, called **SentinelAI**, is an AI-based autonomous security agent designed to assist SOC analysts in monitoring and responding to cybersecurity threats.

The system collects and processes security events from various sources such as network logs, authentication logs, endpoint alerts, and system monitoring data. These events are analyzed using machine learning techniques and rule-based reasoning.

SentinelAI correlates related alerts into meaningful security incidents, identifies suspicious behavior patterns, and assigns risk scores to detected activities.

Based on predefined security policies and contextual analysis, the system can recommend response actions such as alert escalation, endpoint isolation, or access restriction.

By assisting analysts with automated analysis and intelligent recommendations, SentinelAI aims to reduce alert fatigue and improve the efficiency of SOC operations.

2.2 Software Architecture (General)

The proposed architecture integrates SentinelAI with existing SOC monitoring infrastructure.

```
Security Data Sources
(SIEM, IDS/IPS, EDR, IAM, Cloud Logs)
            |
            v
Data Ingestion Layer
(Log collection and normalization)
            |
            v
SentinelAI Agent
------------------------------------
Event Observation Module
```

Alert Correlation Engine
AI Analysis and Reasoning Module
Risk Scoring and Decision Engine
Response Recommendation Module
Learning and Memory Module

```
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             |
             v
```

SOC Analyst Interface / Response System
(Alerting, Incident Handling, Logging)

Component Overview

Data Ingestion Layer
Collects and normalizes logs from multiple security systems.

Correlation Engine
Combines related alerts into meaningful security incidents.

AI Analysis Module
Applies machine learning and behavioral analysis techniques.

Decision Engine
Calculates risk scores and determines recommended actions.

Response Module
Suggests mitigation actions and escalates incidents to analysts.

Learning Module
Improves detection accuracy using feedback from previous incidents.

2.3 Technology and Tools Used

The system will be developed using the following technologies:

- **Python** – main development language
- **Scikit-learn / PyTorch** – machine learning models
- **FastAPI** – backend API service
- **Elastic Stack (ELK)** – log collection and analysis
- **PostgreSQL** – data storage
- **Docker** – containerized deployment

These technologies are widely used in cybersecurity analytics and support scalable data processing and machine learning integration.

3. Implementation Plan

3.1 Main Development Stages

Phase 1 – Requirement Analysis
Study SOC workflows, security logs, and threat detection requirements.

Phase 2 – System Design
Define the architecture and design the SentinelAI modules.

Phase 3 – Development
Implement the AI agent components including event processing and threat detection.

Phase 4 – Testing
Evaluate the system using simulated SOC logs and cybersecurity datasets.

Phase 5 – Documentation
Prepare the final project report and system documentation.

3.2 Expected Schedule

Phase	Timeline
Requirements Analysis	Month 1
System Design	Month 1
Development	Month 2-3
Testing	Month 4
Final Report	Month 5

3.3 Feasibility Assessment

Potential Challenges

• Limited access to real SOC operational data
• Difficulty detecting complex multi-stage attacks

Mitigation Strategies

• Use publicly available cybersecurity datasets
• Simulate SOC scenarios for testing

Required Resources

• Development workstation
• Public cybersecurity datasets (e.g., CICIDS)
• Open-source monitoring tools

These resources are accessible and sufficient for developing a functional prototype.

4. Expected Results

Upon completion, the project is expected to deliver:

• A prototype AI-based security agent for SOC environments
• A system capable of correlating alerts and detecting suspicious activities
• Improved efficiency in analyzing security events compared to manual analysis
• Documentation and architecture design for further system development

5. Evaluation Plan

The proposed system will be evaluated using the following criteria:

Detection Accuracy
Measure the ability of the system to correctly identify suspicious events.

Alert Reduction Rate
Measure the reduction of unnecessary alerts through event correlation.

Incident Response Time
Measure improvements in response time compared to manual analysis.

System Usability
Evaluate how effectively SOC analysts can interact with the system.

Evaluation experiments will be conducted using simulated SOC logs and publicly available cybersecurity datasets.

6. References

[1] É. Lefèvre, "AI-Based Modeling in Software Engineering: Techniques, Applications, and Future Directions," 2022.

[2] I. H. Sarker, "AI-Based Modeling: Techniques, Applications and Research Issues," SN Computer Science, 2022.

[3] D. De Silva et al., "An Artificial Intelligence Life Cycle: From Conception to Production," Machine Learning with Applications, 2022.

✅ **Bản v2.0 này mạnh hơn v1.0 vì:**

- Có **research context rõ hơn**
- Có **research gap**
- Có **evaluation metrics**
- Có **methodology rõ**
- Có **dataset/testing plan**